

__GROUP 12__

No.	NAME	ACCESS NUMBER	CONTRIBUTIONS
:	-----:	-----:	-----:
1.	KAMUGISHA JACKSON	B39090	FOUND THE DATASET
2.	NANTANDA EVE HARRIET PAULINE	B39045	VISUALISATION OF THE DATA
3.	KANSIIME FREDAH	B39123	CLEANED THE DATA
4.	AGWANG LETICIA WINNIEFRED	B40602	EXPLAINED THE LINE GRAPH
5.	AHIMBISIBWE BOSCO	B39926	EXPLAINED THE BOX PLOT
6.	BABIRYE JAELE TENDO	B40149	GAVE AN INITIAL SUGGESTION OF A DATASET FOR INDUSTRIAL PRODUCTION
7.	KASASA ALBERT BWETE	B39110	SUGGESTED WE USE KAGGLE
8.	MUYOMBA ANTONIO	B41617	EXPLAINED THE HISTOGRAM
9.	NSAAWA JONATHAN MUGABI	B40456	CLEAR EXPLANATION OF THE LINE GRAPH & NUMBER OF ROWS TO USE
10.	YAWWE JAYDEN JOHN	B39111	EXPLAINED THE SCATTER DIAGRAM

__WHY WE CHOSE THIS DATASET.__

Its relevant to our current generation, and shows us that students that spend more time on social media platforms are more likely to have a lower GPA than those that spend less time on it.

__WHERE DID WE GET THE DATA__

This particular dataset was sourced from [Kaggle]
<https://www.kaggle.com/datasets/harishyadav0506/impact-of-social-media-on-life>

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
import numpy as np
```

```
data_df=pd.read_csv("Social_media_impact_on_life-selected-columns.csv")
data_df
```

	Age	Gender	Academic_Level	Primary_Platform	Daily_Usage_Hours	\
0	19	Female	Undergraduate	Snapchat	7.5	
1	19	Female	Undergraduate	Instagram	4.6	
2	17	Female	High School	TikTok	10.9	
3	16	Female	High School	YouTube	6.0	
4	17	Male	High School	LinkedIn	3.3	
..	...	...	...	...	...	
190	22	Male	Postgraduate	LinkedIn	7.0	
191	20	Male	Undergraduate	TikTok	2.0	
192	18	Male	Undergraduate	Instagram	2.4	
193	18	Female	Undergraduate	Instagram	5.1	
194	17	Male	High School	TikTok	4.1	

	Device_Type	Sleep_Duration_Hours	Mental_Health_Index	\
0	Smartphone	5.9	73	
1	Smartphone	7.9	90	
2	Smartphone	5.2	70	
3	Tablet	7.6	82	
4	Smartphone	7.6	84	
..	...	...	...	
190	Laptop/PC	4.4	57	
191	Smartphone	7.6	87	
192	Smartphone	6.4	83	
193	Smartphone	7.8	95	
194	Laptop/PC	6.0	90	

	Academic_Performance_GPA
0	2.73
1	3.21
2	3.00
3	3.42

4	3.41
..	...
190	3.32
191	3.45
192	3.41
193	2.89
194	3.46

[195 rows x 9 columns]

HISTOGRAM

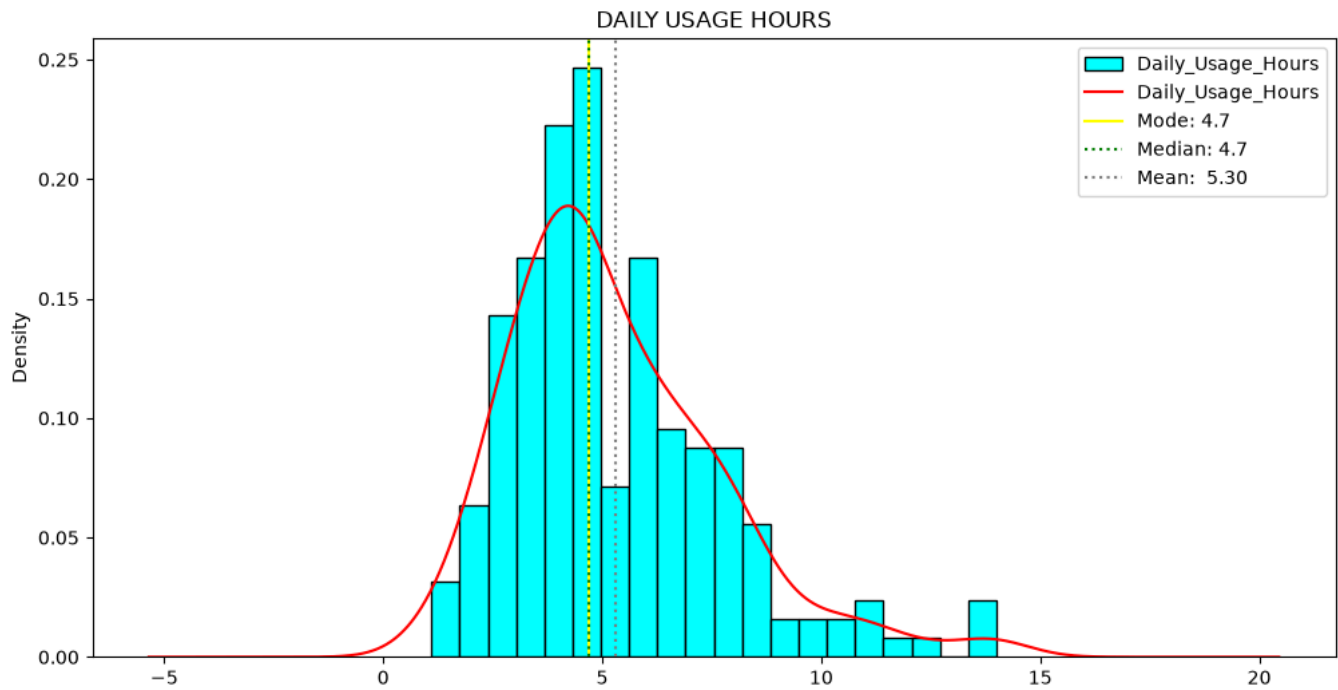
```
mean_value = data_df["Daily_Usage_Hours"].mean()
median_value = data_df["Daily_Usage_Hours"].median()
mode_value = data_df["Daily_Usage_Hours"].mode()[0]

data_df.plot.hist(y="Daily_Usage_Hours", figsize = (12, 6), color = "cyan",bins = 20,edgecolor
= "black", title = "DAILY USAGE HOURS", density = True)
data_df["Daily_Usage_Hours"].plot.kde(color = "Red")

plt.axvline(mode_value, color = "yellow", label = f"Mode: {mode_value}")
plt.axvline(median_value, color = "green", linestyle = "dotted", label = f"Median:
{median_value}")
plt.axvline(mean_value, color = "grey", linestyle = "dotted", label = f"Mean: {mean_value:
.2f}")

plt.legend()
plt.show()
```

<Figure size 1200x600 with 1 Axes>



```
## __POSITIVELY SKEWED__
```

```
### _Explanation_
```

The mean is greater than both the median and the mode.
The kernel Density Estimator tail is stretching to the right.

```
data_df.tail(5)
plt.show()
```

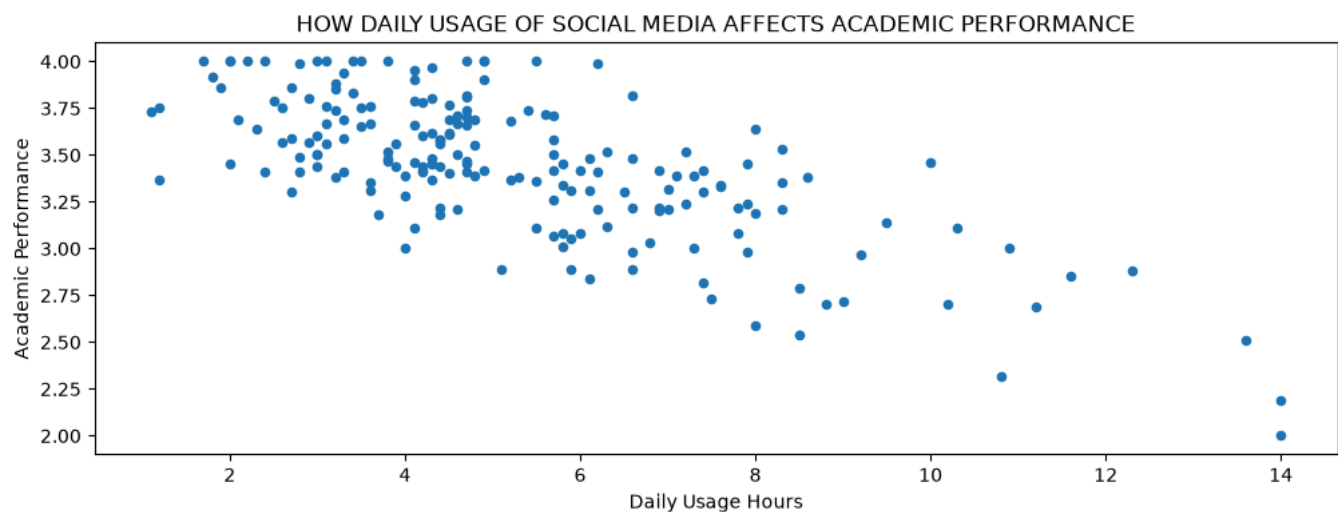
```
# SCATTER DIAGRAM
```

```
data_df.plot.scatter(x = "Daily_Usage_Hours", y = "Academic_Performance_GPA", figsize = (12,
4))

plt.xlabel("Daily Usage Hours")
plt.ylabel("Academic Performance")
plt.title("HOW DAILY USAGE OF SOCIAL MEDIA AFFECTS ACADEMIC PERFORMANCE")
```

```
plt.show()
```

<Figure size 1200x400 with 1 Axes>



Explanation

Most students with high GPAs have a low daily usage of social media while those with a higher daily usage of social media have much lower GPAs.

The scatter diagram shows a strong negative correlation.

```
data_df.plot.scatter(x = "Daily_Usage_Hours", y = "Sleep_Duration_Hours", figsize = (12, 4))
```

```
x = data_df["Daily_Usage_Hours"]
```

```
y = data_df["Sleep_Duration_Hours"]
```

```
m, b = np.polyfit(x, y, 1)
```

```
plt.plot(x, m*x + b, color='red')
```

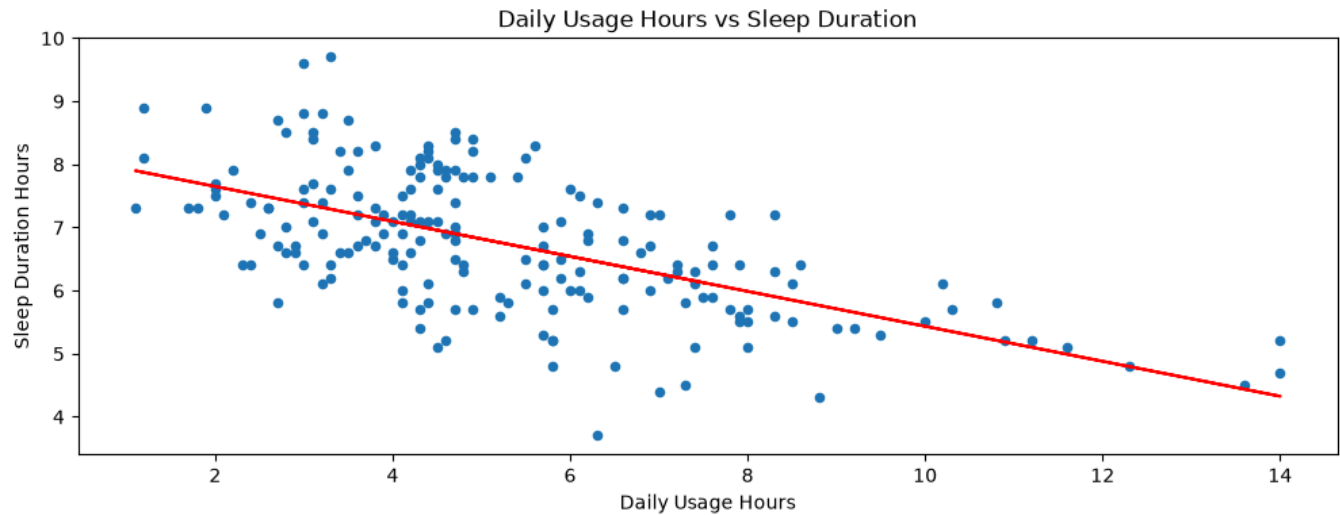
```
plt.xlabel("Daily Usage Hours")
```

```
plt.ylabel("Sleep Duration Hours")
```

```
plt.title("Daily Usage Hours vs Sleep Duration")
```

```
plt.show()
```

<Figure size 1200x400 with 1 Axes>



```
## __COMMENT__
```

The scatter diagram shows that students that spend more time on social media, spend less time sleeping.

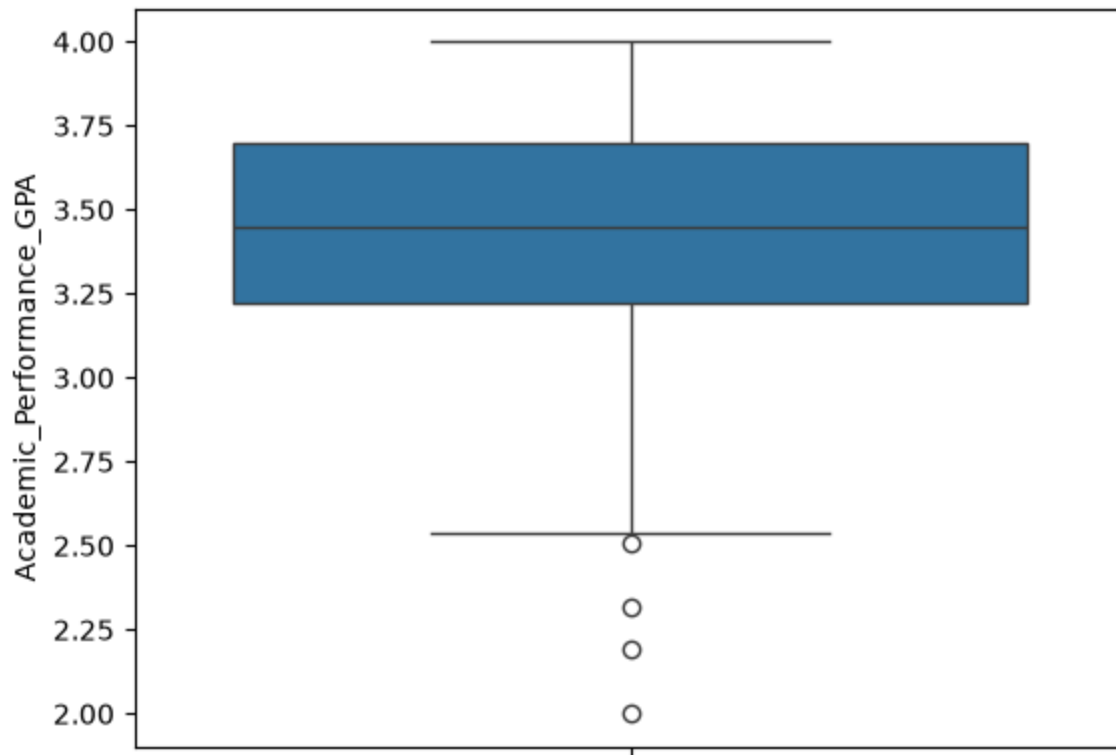
The scatter diagram shows a strong negative correlation.

```
# BOX PLOT
```

```
sns.boxplot(data_df["Academic_Performance_GPA"])
```

```
plt.show()
```

<Figure size 640x480 with 1 Axes>



__EXPLANATION__

This box plot shows the distribution of students' Academic Performance (GPA).

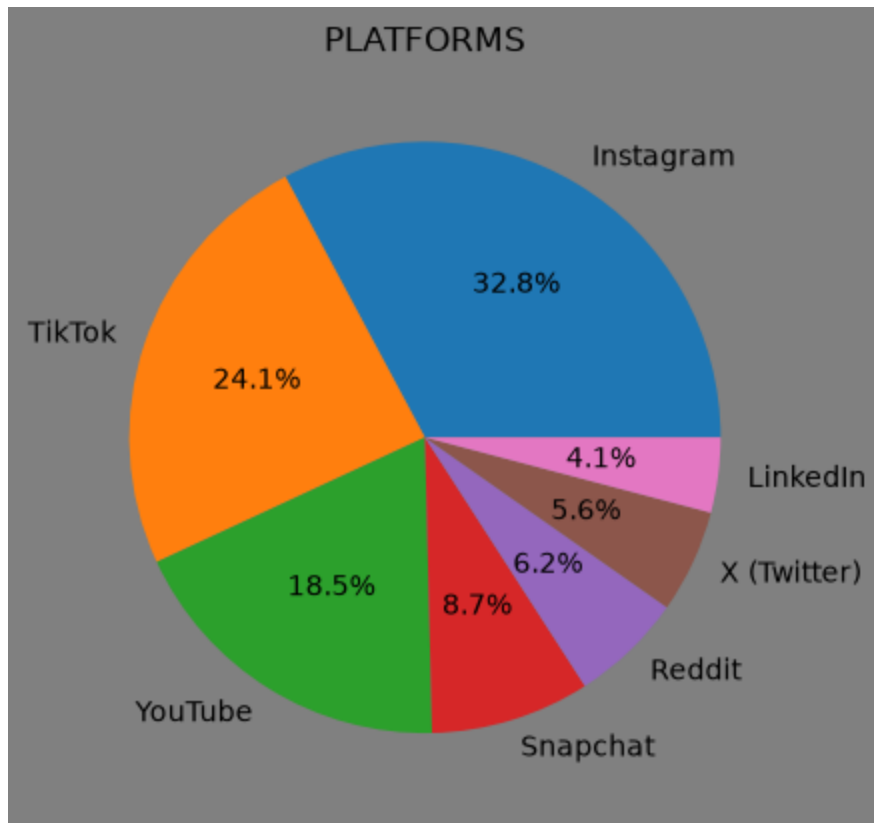
- * The middle line in the box is the median GPA, which is about 3.45.
- * The box shows where most students' GPAs are found, roughly between 3.25 and 3.70.
- * The whiskers show the wider range of normal GPA values, from about 2.65 to 4.00.
- * The outliers show that few students have unusually low GPAs, around 2.0–2.5.
- * Overall, the plot shows that most students are concentrated around 3.25–3.70.

__PIE CHART__

```
data_df['Primary_Platform'].value_counts().plot.pie(  
    autopct = '%1.1f%%'  
)
```

```
plt.gcf().set_facecolor("gray")  
plt.title("PLATFORMS")  
plt.ylabel('')  
plt.show()
```

<Figure size 640x480 with 1 Axes>



__COMMENT__

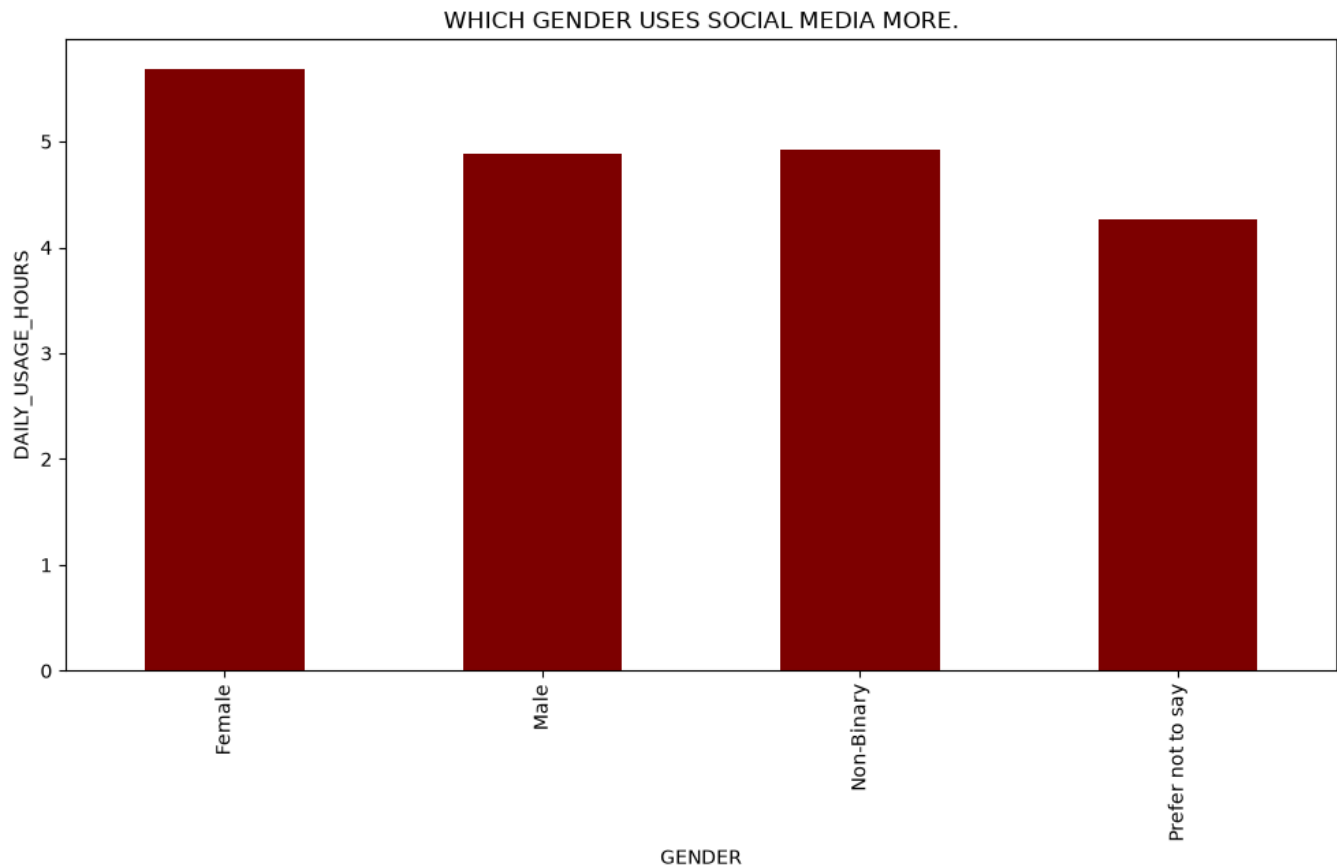
This pie chart shows the different social media platforms used by students and the percentage of students using each platform.

__BAR GRAPH__

```
data_df.groupby('Gender')['Daily_Usage_Hours'].mean().plot.bar(figsize = (12, 6), color =  
"maroon")
```

```
plt.title("WHICH GENDER USES SOCIAL MEDIA MORE.")
plt.xlabel('GENDER')
plt.ylabel('DAILY_USAGE_HOURS')
plt.show()
```

<Figure size 1200x600 with 1 Axes>



__COMMENT__

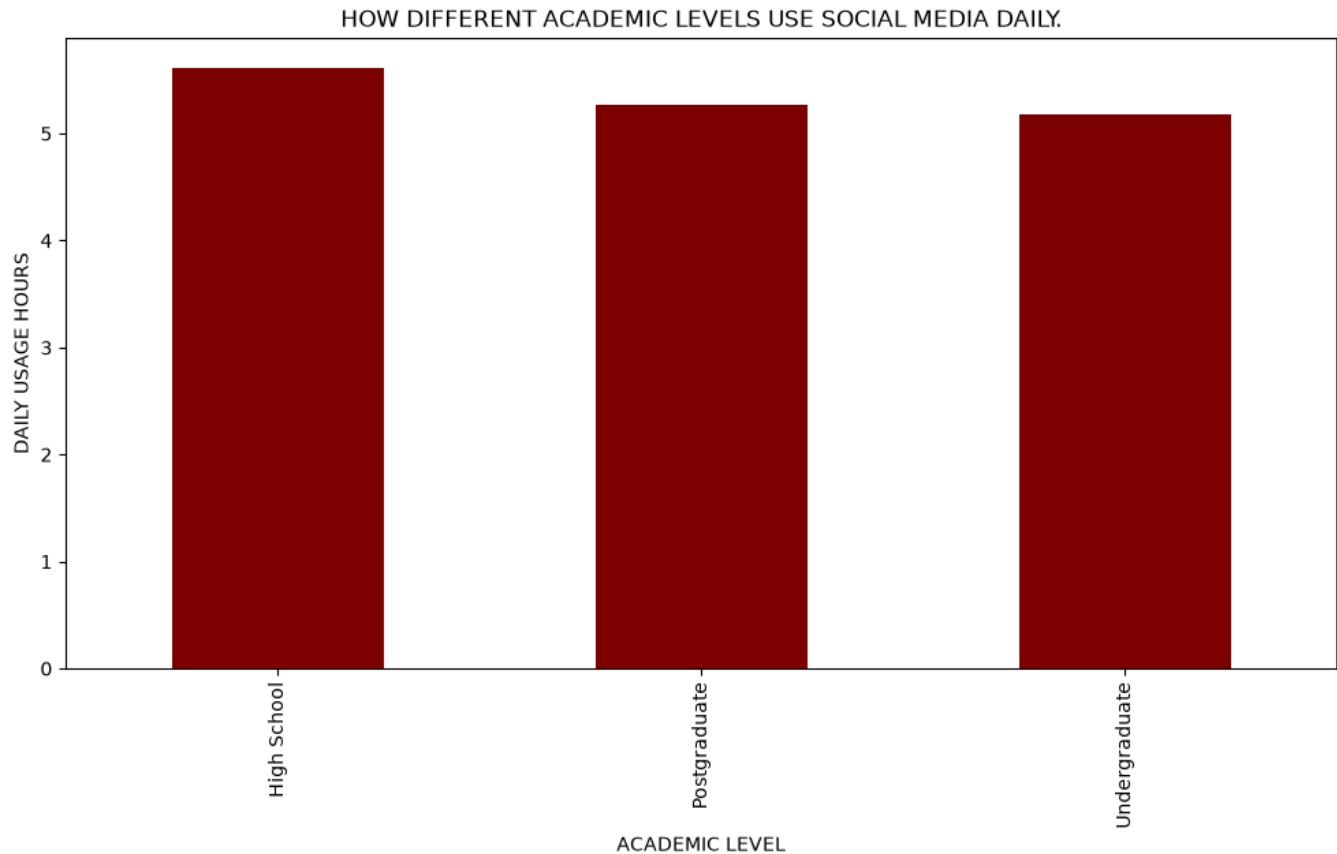
The bar graph shows that generally females spend the most time using social media than other gender groups.

```
data_df.groupby('Academic_Level')['Daily_Usage_Hours'].mean().plot.bar(figsize = (12, 6),
color = "maroon")
```

```
plt.title("HOW DIFFERENT ACADEMIC LEVELS USE SOCIAL MEDIA DAILY.")
```

```
plt.ylabel('DAILY USAGE HOURS')  
plt.xlabel('ACADEMIC LEVEL')  
plt.show()
```

<Figure size 1200x600 with 1 Axes>



__COMMENT__

The bar graph shows that students in highschool spend more time on social media than other academic levels.

__WHAT CONCLUSION CAN WE GET FROM THE DATASET?__

We conclude that spending a lot of time on *Social media* might lower our GPA or might increase it

From the statistics and visualizations made we can see most students who spend a lot of time on social media perform badly, although a few perform even better than those who spend few

hours on it.

- The possibility for those who spend a lot of time on social media but perform well is that they might be acquiring knowledge from the social media, while those who perform poorly might be using it for non-academic purposes.

* From the line graph, we deduced that most students use social media platforms for an average duration (4-9 hrs.)